MILITARY SPECIFICATION

INTRAVENOUS INJECTION SET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, disposable intravenous injection sets (infusion apparatus) designed for use in the administration of parenteral solutions from standard plastic and glass solution containers, without the use of an airway cannula.

1.2 Classification. Intravenous injection sets shall be of the following types as specified (see 6.2):

Type I. - Regular set (20 drops/ml.)

Type II. - Pediatric set (60 drops/ml.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-N-196	Needles, Hypodermic, for Luer-Syringes.
MMM-A-250	Adhesive, Water Resistant, for Sealing Fiberboard Boxes.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
Bureau of Biologics. Food and Drug Administration
Code of Federal Regulation Part 273. Minimum Requirements (as applicable).

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockville, Maryland 20852.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI Z70.1 (UDC615.47)	American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber and plastic components. All rubber and plastic components shall be of the best quality and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterilizable, free from pyrogens, compatible, and safe when tested as specified in section 4. Those rubber and plastic components coming in contact with solutions being administered, and end protectors, shall contain not more than 5 parts per million (p.p.m.) total of lead, barium, and cadmium when tested as specified in 4.4.1, or 4.4.1.1. The plastics shall meet the requirements of the Pharmacopeia of the United States (USP) for Class II plastics and for transfusion and infusion assemblies when tested as specified in 4.4.2.

3.1.1.1 Plastic tubing. Plastic tubing shall be clear, colorless, transparent, flexible polyvinyl chloride, having a hardness of 75 (plus or minus 5) Shore A durometer, when tested as specified in 4.4.3. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be clean and free of particles and shall be free from all liquid lubricating substances which oxidize, exude, or sweat-out to the surface of the tubing.

3.1.2 Needles.

3.1.2.1 Cannulae of needles. Cannulae of needles shall be fabricated from 304 AISI stainless steel, and shall meet the corrosion-resistance, elasticity, and resistance-to-breakage tests as specified in GG-N-196. The cannulae shall be fabricated from either seamless drawn, or welded and drawn tubing.

3.1.2.2 Hubs. Hubs shall be fabricated from a suitable plastic (containing no exudable toxic materials), aluminum alloy, or nickel-plated brass. Aluminum hubs shall be degreased and then either anodized, or passivated with nitric acid solution resulting in a matte, nontinted finish.

3.1.3 Other material. All other material hereinafter specified shall be of the best quality and suitable in all respects for the purpose intended.

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3.2 Type I - Regular set. The type I set shall be a sterile, nonpyrogenic, safe (nontoxic), disposable intravenous injection set (infusion device) for use in the administration of parenteral solutions from standard plastic and glass solution containers (see 3.2.2). The style and design shall be in accordance with Figure 1 and as specified herein. The set shall have a cap with piercing device, a flexible plastic calibrated drip chamber, an air vent with a check valve covered by a bacterial filter, one female needle adapter, a length of plastic tubing, an injection site, a male needle adapter, an intravenous needle, a flow control device, and end protectors. The drip chamber shall be molded of suitable, flexible, colorless clear plastic. The base of the cap shall be permanently attached to the drip chamber in a leakproof manner and shall not be solvent welded or cemented. The assembled drip chamber shall be so shaped as to provide a convenient hand grip for the insertion of the piercing device into the stopper or port of the solution container. The assembly of the set shall be structurally sound and shall not leak air or fluid during use. Open ends of the set shall be tightly covered (sealed) with readily removable covers, or end protectors, which shall maintain the sterility and freedom from pyrogens of the fluid path after sterilization.

3.2.1 Drip chamber. The drip chamber housing shall have a volume between 4 ml minimum and 5.5 ml maximum.

3.2.2 Cap (piercing device). The cap shall consist of an integrally molded unit including a piercing device (spike), and an air inlet with built in check valve. The piercing device shall be suitable rigid plastic, dual-channeled, conical spike, one channel of which shall serve as an air inlet, the other channel being extended to form the drip orifice. The piercing device shall be not less than 7/8 inch in length, with a diameter of nominal 1/4 inch at the base (proximal end). The piercing device tip shall be sharpened to such a point as to be suitable for puncture of, and insertion into, the stopper or port of a standard solution container (as specified herein) without undue pressure (see 3.2.2.1). Edges of fluid and air inlet passage openings shall be rounded or dulled to prevent stopper coring. The piercing device shall be ruggedly designed to prevent coring or shearing, and shall be structurally sound. It shall be firmly retained in a leakproof manner when inserted in the stopper or port of a standard solution container. The piercing device shall not break or leak air or fluid during use. It shall not be brittle or soft, or become brittle or soft under normal storage conditions. The standard solution container stopper referenced above shall be in accordance with requirements contained in the Defense Medical Purchase Description for NSN 6505-00-116-4600. The passages in the piercing device shall be properly designed to prevent air from passing directly from the air passage to the fluid passage under maximum flow conditions. The air inlet passage shall contain a valve seat and a stainless steel check ball and vent the solution container through the bacterial filter (without the need

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of an airway cannula) during administration of solution. In addition, the air inlet shall provide a means of adding fluids either by syringe or by series hookup of parenteral solution equipment through a passage extending down the stem of the spike to an opening on the side of the housing cap. This opening shall have a female Luer taper in accordance with Figure 3 of ANSI Standard Z70.1. It shall be capable of accepting in a leakproof manner, a standard male luer lock syringe, or the needle adapter of the series hook-up equipment with the spike properly inserted in the stopper, or port, of a standard solution container. The air inlet cavity shall be tested for diameter and depth, as specified in 4.4.4, and shall fit the plug gage within the "GO" - "NO GO" limits of the feeler gage as shown on Figure 5 of ANSI Standard Z70.1. The air inlet shall not leak when tested as specified in 4.4.8. This air inlet opening shall be capped with a removable and replaceable porous air filter which shall prevent the ingress of bacteria (see 3.1.2.2), and which shall serve as an air filter during storage and use. The filter cap shall form a snug fit with, and shall be retained on, the opening until manually removed. The fluid passage in the spike shall extend into the drip chamber, and shall be calibrated to deliver 15-20 drops per milliliter (ml.) when tested as specified in 4.4.5. Shall maintain the 15-20 drops per milliliter (ml.) when tested as specified in 4.4.5. Shall maintain the 15-20 drops per ml drop size during administration of intravenous solutions for a minimum of 48 hours.

3.2.2.1 Durability of piercing device. When tested as specified in 4.4.6 the device, or the cap, shall not loosen or permit leakage, nor the device or cap become damaged. The piercing device shall be capable of being inserted into the standard solution containers, as described in 4.4.6, without difficulty, the tip (or point) of the device not breaking or bending to the degree of causing difficulty in insertion into standard solution containers.

3.2.2.2 Air inletting bacterial filter. The air inletting bacterial filter shall be of such design and material as to comply with the following performance requirements when tested as specified in 4.4.7:

(a) Bacterial retention. The filter shall be such that the rate of contamination of the test media shall be not greater than 0.1% (1-1000).

(b) Air passage. The filter shall pass air at a minimum rate of 150 ml. per minute.

(c) Resistance to water. The filter shall support a 15-inch static head of water for 10 seconds with no wetting of the filter element.

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3.2.3 Plastic tubing. The plastic tubing shall be fabricated of material as specified in 3.1.1.1 and shall be a minimum of 78 inches. It shall be firmly and permanently secured, in a leakproof manner, to the drip chamber at one end and to a male needle adapter at the other end in such a manner as to form an airtight, leakproof connection which shall not separate when tested as specified in 4.4.8. A "Y" type injection site fabricated of clear or translucent rigid plastic shall be connected to the tubing at a nominal distance of six inches above the needle adapter. Two of the legs shall be in line, and the third leg at a nominal 30° angle. Top leg of inline portion of "Y" shall have a standard Luer taper into which a resealable diaphragm is securely fitted. Tubing is fastened to other two legs with single leg toward the needle adapter.

3.2.4 Rubber diaphragm. The rubber diaphragm on the medicinal entry assembly leg shall be molded from natural, non-oxidizing, low-sulfur, wax free rubber of 45 ± 5° Shore A Durometer.

3.2.5 Control clamp. Control clamp shall be fabricated of plastic, a screw-type clamp from which the screw cannot be accidentally removed. The clamp shall be supplied on tubing in open position and shall provide a constant setting and quick shut off from the operating position, shall be movable, and shall be capable of one-handed operation. To assure constant delivery of fluid, it shall be impossible for the clamp setting to be changed drastically by an accidental jerk on the tubing. When tested as specified in 4.4.10, the clamp shall be capable of accomplishing complete quick shutoff (i.e. no falling drop in drip chamber or from needle adapter (3.2.6), shall regulate flow of the solution from complete shutoff to maximum flow, and with a constant head of water shall maintain a constant flow rate at any point between complete shutoff and maximum flow. In addition, when the clamp is operated from its fully open position to complete shutoff, and this operation is repeated ten times while the clamp is in the same position on the tubing, the tubing shall not be cut, pierced, or otherwise damaged so as to cause leakage or restriction of flow of solution.

3.2.6 Male needle adapter. The male needle adapter shall be fabricated of suitable plastic and shall be tapered to fit a needle having a standard Luer hub in accordance with ANSI Standard Z70.1 in a tight leakproof manner (see 3.8). The needle adapter shall not separate from the tubing or show evidence of leakage when tested as specified in 4.4.8.

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3.2.7 Intravenous needle. The needle shall be a 19 or 20 gage, 1-1/2 inch cannula intravenous regular bevel needle. Except as specified herein, the needle shall be in accordance with requirements of GG-N-196 for type I, point style a, hub style 1 or 2. The cannula and hub shall be fabricated of material as specified in 3.1.2.1 and 3.1.2.2, respectively. The hub cavity shall be constructed in accordance with Figure 3 of ANSI Standard Z70.1. The needle shall fit the needle adapter in such a manner as to make a tight, leakproof connection. The hub cavity shall be tested for diameter and depth and fluid tightness as specified in 4.4.11. Hubs shall fit the plug gage within the "GO" - "NO GO" limits of the feeler gage as shown on Figure 5 and 6 of ANSI Standard Z70.1. Hubs which do not fit the plug gage within these limits shall be considered not acceptable. In addition, when the hub is fitted onto the plug gage as specified in 4.4.11.1, there shall be no movement of the hub on the gage when the cannula is pressed lightly from side to side in all directions. When tested for fluid tightness as specified in 4.4.11.2 the swaged connection of the cannula with the hub, and the connection of the hub with the test gage, shall not show leakage exceeding the formation of a falling drop. The needle shall be supplied with hub attached to the tapered end of the needle adapter, or in close approximation for such attachment. The needle shall be covered by the end protector in such a manner as to be readily and easily attached to the needle adapter and retain its sterility. The needle marking required by GG-N-196 may be omitted.

3.2.8 End protector. The protector shall be attached to the needle adapter while the needle is contained within the protector in a manner which will prevent contact of the needle point with the inside end of the protector and also provide a loose fit over the Luer end of the needle adapter. End protector shall be readily removable, transparent or semi-transparent plastic tube, sheath, or molded cap constructed in such a manner that when placed over each open end of the intravenous injection set it shall, after sterilization, maintain sterility, freedom from pyrogens, immobility of the needle, and integrity of the entire fluid path of the infusion device until manually removed.

3.3 Type II - Pediatric set. Type II pediatric intravenous injection set shall comply with the requirements specified herein for the type I (Regular) set (see 3.2) except that the fluid passage in the spike shall be calibrated to deliver 60 ± 9 drops per milliliter when tested as specified in 4.4.5, and shall be supplied without a needle. Shall maintain the 60 drop per milliliter drop size during administration of intravenous solutions for a minimum of 48 hours at approximately 60 drops per minute. The plastic tubing shall be as specified in 3.2.3 except shall be a minimum of 71 inches in length.

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3.4 Manufacturer's record. The manufacturer shall place a lot control number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made available for inspection by any duly qualified medical officer of the Armed Forces, U. S. Department of Health, Education and Welfare, any official of the Federal Food and Drug Administration, or any duly authorized Government representative. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent record of the manufacturer.

3.5 Sterilization, safety, and freedom from pyrogens. All equipment shall be safe. Fluid paths, end connectors, and needles of type I sets and fluid paths, and end connectors of type II sets shall be sterile and pyrogen-free. All the above shall be in accordance with current revisions of the minimum requirements of the National Institutes of Health for citrated whole blood. The fluid path, end connectors, and needles shall be pyrogen-free when tested as specified in 4.4.12 and sterile when tested as specified in 4.4.13. The sets shall be safe when tested for safety as specified in 4.4.14.

3.6 Certificates. Certificates of sterility, freedom from pyrogens, safety, and bacterial retention as specified below, shall be submitted. For destination inspection, certificates shall be submitted by the contractor directly to Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101, Attention: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection certificates shall be furnished to the procuring activity via the government inspector.

3.6.1 Certificate of sterility. A certificate of sterility, listing all sterility lot or control numbers, shall state that samples from each sterility lot or control have been tested for sterility, as required in 3.5 and were found to be sterile. The certificate shall reference the method used in determining sterility.

3.6.2 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.5 and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

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3.6.3 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.5, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.7 Compatibility. The materials of which the type I and type II sets are manufactured shall not react with, or be acted upon by, the liquids with which they will be in contact during normal use and operation, nor shall there by any materials incorporated in the set which will cause any deleterious effect when the set is used for the purpose intended.

3.8 Leakage and separation of set. Type I and type II sets shall be tested for leakage and separation of set as specified in 4.4.8, and 4.4.8.1. Sets which show leakage or separation at any part of the set shall be rejected. A leak is defined as the formation of an air (gas) bubble or liquid as appropriate, which breaks away from the set within the specified test time (5 seconds).

3.9 Solutions flow rate. When tested as specified in 4.4.15, the intravenous injection set, with the filter in its normal dry condition shall permit the passage of 100 cc of fluid in not more than 75 seconds (for Type I) or 5-1/2 minutes (for Type II), and not more than 120 seconds for (Type I) or 8-1/2 minutes (for Type II) with the filter wet (immerse filter in distilled water at room temperature for at least 10 minutes).

3.10 Instructions. Individual set carton shall have detailed instructions for proper use of the set for single use. Detailed instruction sheet for tandem use shall be inserted in each unit package (48's).

3.11 Workmanship. Workmanship shall be first class throughout. The intravenous injection sets shall be free from defects which detract from their appearance or impair their serviceability.

3.12 Government furnished material. The Government shall furnish the contractor the dextrose injection bags (NSN 6505-00-083-6538) needed to perform the leakage and drip test specified in 4.4.8.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the components or materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Lot

4.2.1.1 Plastic. For purposes of this specification, a lot or batch shall consist of a quantity of plastic moldings or tubing of uniform composition produced under like conditions and shall represent one continuous injection mold or extrusion run of plastic.

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4.2.2 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination shall be one intravenous injection set.

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.3 For tests.

4.2.3.1 For sterility test. Sampling for sterility test shall be in accordance with the United States Pharmacopeia (USP).

4.2.3.2 For pyrogen, and safety tests. Sampling procedures for pyrogen and safety tests shall be conducted as indicated in the USP, for transfusion and infusion assemblies.

4.2.3.3 For extractable metals test. Samples shall be taken from each lot or rubber or plastic as indicated in 4.4.

4.2.4 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.

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Table II. Sampling for Other Tests

Component (unit of product)	Characteristic	Req.	Test procedure	Insp. level	AQL (percent defective)
Plastic and rubber components	Heavy metal (extractable)	3.1.1	4.4.1	**	*
	Biological	3.1.1	4.4.2	**	*
	Tubing hardness	3.1.1.1	4.4.3	****	*
1-Intravenous injection set	Dimensions	3.2	——	S-3	*
1-Intravenous injection set	Air inlet female luer taper	3.2.2	4.4.4	S-3	*
1-Intravenous injection set	Drop test	3.2.2 and 3.3	4.4.5	S-3	*
1-Intravenous injection set	Piercing device durability	3.2.2.1	4.4.6	S-3	*
1-Intravenous injection set	Leakage, and separation of set	3.2.2, 3.2.6 and 3.8	4.4.8 and 4.4.8.1	I, except sample size shall be a minimum of 125 samples	*
1-Intravenous injection set	Reseal (rubber diaphragm)	3.2.4	4.4.9	S-3	*
1-Intravenous injection set	Control clamp	3.2.5	4.4.10	S-3	*
1-Intravenous injection set	Intravenous needle	3.2.7	4.4.11	S-3	*
1-Intravenous injection set	Freedom from pyrogen	3.5	4.4.12	***	*
1-Intravenous injection set	Sterility	3.5	4.4.13	***	*

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Table II. Sampling for Other Tests (Cont'd)

Component (unit of product)	Characteristic	Req.	Test procedure	Insp. level	AQL (percent defective)
1-Intravenous injection set	Safety	3.5	4.4.14	***	*
1-Intravenous injection set	Solution flow rate	3.9	4.4.15	S-3	*

* Acceptance number shall be zero for all sample sizes.

** Each lot of product need not be biologically tested provided evidence is furnished that samples of the formulation and identical method of processing had previously been tested, that they comply with USP requirements for Class II plastics and that they are the same as being offered on this contract.

*** Sampling for tests shall be on component parts.
**** Sufficient sample units to perform the test shall be selected from each lot.

4.2.5 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.2.6 For plastic tubing hardness test. Sufficient samples shall be selected from each lot to perform the test. Failure to comply with test requirements shall be cause for rejection of the lot represented.

4.2.7 For bacterial retention test. This test shall be performed at least once for each contract. The samples selected shall represent the exact composition, treatment, and construction to be used throughout the entire contract. Any change in the filter shall require additional testing. The number of samples specified in 4.4.7.1 shall be randomly selected from production lots of filters used. Failure of any sample to comply shall be cause for rejection of the filters represented.

4.3 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Examination, however, shall not be restricted to the classified possible defects listed in Table III.

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Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	End protectors missing or dislodged.
102	Cotton missing (or inadequate amount) from open type protector.
103	Not free of bloom or tackiness.
104	Set incomplete.
105	Style and design not as specified.
106	Presence of foreign matter.
107	Set not free of holes or cuts.
Minor	
201	Packaging, packing and marking not as specified.
202	Instructions missing, incomplete or illegible.
203	Contractor's name or registered trademark, missing, incomplete, or illegible.

4.4 Tests

4.4.1 Heavy metals extractable test. A 2 gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small slivers, so that no thickness exceeds 3 mm., and eluted for 72 hours at 50° C., 24 hours at 70° C., or 60 minutes at 121° C. in one menstrua: 0.9% sodium chloride. Each gram of the material shall be covered with 5 milliliters (ml.) of the menstruum. The menstrua are collected, agitated after elution, and then submitted to spectographic atomic absorbtion spectrophotometric or specific wet analysis for lead, barium, and cadmium. The metals are reported in parts per million. Thus 1.0 ppm in elute is equivalent to extracting 1.0 mg. from 200 g of test material dispersed in 1,000 ml. of elution menstruum.

4.4.1.1 Alternate heavy metals extractable test. Accurately weigh a 5 - gram sample for each of the metals (barium, cadmium or lead) into a 1 - oz. Vycor crucible. Reduce to ash in a muffle furnace at 650° C for eight hours. Let cool to room temperature and add 15 ml. of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minute, remove, let cool, and filter through Whatman #2 filter paper into a 50 - ml. volumetric flask. Rinse the residue on the filter with deionised water and dilute to 50 ml. Prepare standard solutions of 1, 2, and 5 ppm for each of the metals, namely barium, cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a Hollow Cathode Ba, Cd, or Pb lamp for each of the respective metals and a 10X scale expansion on a spectrophotometer,

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such as the Perkin-Elmer Model 303, Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm barium, cadmium, or lead}$$

A = Absorbance
F = Factor = $\frac{(\text{Concentration standard in ppm Ba, Cd, or Pb})^*}{\text{absorbance}}$

W = Sample weight in grams
*The three standards are averaged to give ppm/unit A.

4.4.2 Biological tests. The plastics shall be tested in accordance with the USP for Class II plastics.

4.4.3 Tubing hardness. Plastic tubing hardness shall be determined in accordance with method 3021 of Federal Test Method Standard No. 601. The test shall be conducted at a room temperature of 27° ± 5°C (80° ± 90° F) on samples which had been conditioned at this temperature for not less than 3 hours prior to testing. The test sample shall be laminated to a thickness of not less than 1/4 inch. The gage reading shall be made 15 seconds after application of the instrument.

4.4.4 Air inlet female Luer taper. The air inlet female Luer taper shall be tested on the male Luer taper gage in accordance with plug gage as shown on Figure 5 of ANSI Standard Z70.1

4.4.5 Drop test. The intravenous injection set shall be tested at room temperature, using water at a rate of 100 ± 5 drops per minute. The number of drops required to deliver 2 ml. of water shall be counted, and the drops per ml. calculated from this value.

4.4.6 Piercing device durability test. Samples of finished sets, after sterilization, shall be used for this test. The piercing device (on assembled set) shall be inserted into, and removed from, a standard solution container (specified in 3.2.2) in the normal manner. This operation shall be repeated twice. The piercing device shall then be immersed in a solution of 5 percent dextrose in isotonic sodium chloride solution (in accordance with requirements of the USP monograph for "Dextrose and Sodium Chloride Injections") for 4 hours. The piercing device shall then be removed from the solution and immediately inserted into a standard solution container as specified in 3.2.2.

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4.4.7 Air inletting bacterial filter.

4.4.7.1 Bacterial retention. The following test shall be performed on not less than 100 sample filters randomly selected to represent specific materials used in production batches of filters, and four additional test bottles shall be used for control test for each test period: Detach drip chamber assembly from each sample sterile intravenous injection set by sealing and cutting the plastic tube within 2 inches of drip chamber so that the tubing interior remains sterile. Regular parenteral 1000 ml. (1 liter) bottles having a standard closure shall be filled with 50 ml of fluid thioglycollate medium, USP, evacuated to at least 25 inches of mercury (hg), sealed and sterilized. Introduce a sufficient concentration of A. aerogenes into a number 44 De Vilbiss pocket nebulizer, or similar apparatus. Connect nebulizer to air filter on a drip chamber assembly by means of gum rubber tubing with necessary reducers and connectors. Adjust the concentration of organisms in the nebulizer to obtain a minimum of 300 organisms in the air volume replacing the vacuum in the liter test bottle. Insert the piercing device of each prepared sample drip chamber into a bottle closure in the normal manner. When the vacuum in the liter test bottle has been displaced, disconnect the testing apparatus from the filter and shake the bottle thoroughly to mix the aerosol content and to wet all surfaces. Retain the drip chamber assembly in the bottle for venting and incubate at 31° ± 1° C for seven days. As a control to assure that the aerosol is challenging the filter correctly, remove the air filter from the drip chamber assembly of two additional bottles at the start and two additional bottles at the end of each and every test period and connect the nebulizer. When the vacuum has been displaced, disconnect the nebulizer and replace the filter. After mixing the bottle contents, determine the bacterial count of the bottle. The rate of contamination shall be as specified in 3.2.2.2.

4.4.7.2 Air passage. The air-inletting bacterial filter shall pass air as specified in 3.2.2.2, when tested assembled in the filter cap and under a pressure equal to a column of 16 inches of water. The flow shall be measured with a Brooks Rotometer, or equivalent apparatus.

4.4.7.3 Resistance to water. The air-inletting bacterial filter shall be tested for resistance to water in its filter cap by placing the filter cap on a fitting connected to a 15-inch column of water. The filter shall be examined for wetting at the end of 10 seconds exposure.

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4.4.8 Leakage and separation of set. The test shall be applied to those samples subjected to the piercing device durability test (see 4.4.6). The needle, and the cap containing the air inletting bacterial filter shall be removed. The plastic needle adapter of the set shall be fitted into the cavity of the air inlet. The set shall then be immersed in water and subjected to a pneumatic pressure of 8 pounds per square inch gage (psig) for 5 seconds. The pressure shall be applied through the piercing device and the set observed for separation and leakage.

4.4.8.1 Leakage and drip test. An infusion rate of 40 ± 5 drops per minute shall be established on the intravenous injection set when set-up as shown in Figure 2. The solution bag shall be squeezed by hand to 100 mm ± 10 of Hg maintained for 5 seconds, observed for leakage after which the infusion rate shall be measured which shall not vary more than 10 drops per minute from the original infusion rate setting.

4.4.9 Reseal test. The rubber diaphragm shall reseal without a falling drop within 10 seconds after puncture with a nonlubricated 18 gage needle when subjected to an internal pressure (hydrostatic pressure) of 4 p.s.i.g. for 30 seconds.

4.4.10 Clamping device performance test. The assembled set (except for needle) shall be connected, in the normal manner, to a standard solution container as specified in 3.2.2 except, as an alternate, the container may be filled with saline solution or water. The container, with set connected thereto, shall be suspended, in the normal manner, at a sufficient height to permit the entire length of the tubing of the set to hang vertically. The ability of the clamp to accomplish complete shut off, to regulate flow, and with a constant head of water maintain a constant rate of flow, shall be tested with the clamp at a point on the plastic tubing within two inches above the molded rubber part and shall meet the requirement specified in 3.2.5. The tests shall be performed with the tubing hanging vertically at all times. The rate of flow shall be determined either by counting the drops in the drip chamber during a measured time period, or by recording the time required to collect a measured volume of fluid from the needle adapter.

4.4.11 Intravenous needle.

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4.4.11.1 Diameter and depth of hub cavity. The needle hub cavity shall be tested on the male Luer taper gage in accordance with the plug gage of Figures 5 and 6 of ANSI Standard Z70.1. (This gage shall be as shown in ANSI Standard Z70.1, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.11.2). The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with Figures 5 and 6 of ANSI Standard Z70.1.

4.4.11.2 Fluid tightness. With the needle in horizontal position, the hub shall be attached by means of a plug gage, as shown on Figure 5 of ANSI Standard Z70.1 (except that gage shall have a fluid path), to a water pressure source having a pressure gage. The water shall be run through the needle so as to fill any hub cavity space and the lumen of the cannula; the lumen then sealed by a piece of solid rubber slipped on to the point, and a pressure of 45 pounds per square inch applied for 10 seconds.

4.4.12 Freedom from pyrogens. Samples of the finished sets shall be tested for freedom from pyrogens, in accordance with the USP, for transfusion and infusion Assemblies.

4.4.13 Sterility. Samples of the finished sets shall be tested for sterility in accordance with the procedures for parenteral devices as described in the USP.

4.4.14 Safety test. The finished sets shall be tested for safety in accordance with the USP, for transfusion and infusion assemblies.

4.4.15 Solution flow rate. The finished intravenous injection set shall be connected to a standard solution container as specified in 3.2.2 in the normal manner, the entire assembly inverted and suspended so that there is a 30 inch column of fluid in the set and container and no restriction of flow imposed by the needle or the clamp. The time necessary for passage of 100 cc. of fluid through the entire set shall be determined. This test procedure shall be repeated after purposely wetting the filter with water as specified in 3.9.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing forty-eight (48) sets, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not Required	4 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior containers of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Individual box. Each set shall be packaged in a sealed box. At the option of the contractor, each box may have a die-cut window and overwrapped in clear plastic or cellophane. Wrapper of box shall be sealed to prevent accidental opening. Wrapper shall have a tear strip for ease of opening.

5.1.3.2 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. As an alternate, boxes shall be closed by gluing top and bottom flap with adhesive conforming to MMM-A-250. Metal stitches are not permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirement of applicable table shall apply for PPP-B-636 box.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall close by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual box. Each box shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act. Each box shall contain instructions as specified in 3.10, and the legend: "FLUID PATH AND END CONNECTORS ARE STERILE AND PYROGEN FREE" or similar legend.

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Marking shall include the lot (control) number, contract or purchase order number, and name of contractor. When the contractor is not the manufacturer, name of manufacturer shall also appear.

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In addition, marking shall include the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat Sterilization) which shall be shown following the level of packaging and date of packaging. Example: "A-2/70-G". Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. In addition, the method of sterilization as specified in 5.3.2 shall be shown.

NOTE: Each lot number shall be traceable to a specific sterilizer charge.

6. NOTES

6.1 Intended use. The intravenous injection set is intended for accurate administration of parenteral solutions from standard plastic and glass solution containers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

NSN 6515-00-117-9021	INTRAVENOUS INJECTION SET, with Flexible Drip Chamber, 19 or 20 Gage Needle, Disposable, 48s (Type I).
NSN 6515-00-118-1341	INTRAVENOUS INJECTION SET, With Flexible Drip Chamber, Without Needle, Pediatric, Disposable, 48s (Type II).

This specification does not include all types, sizes, or grades as indicated by the title or all types which are commercially available.

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Custodians:

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Review activities:

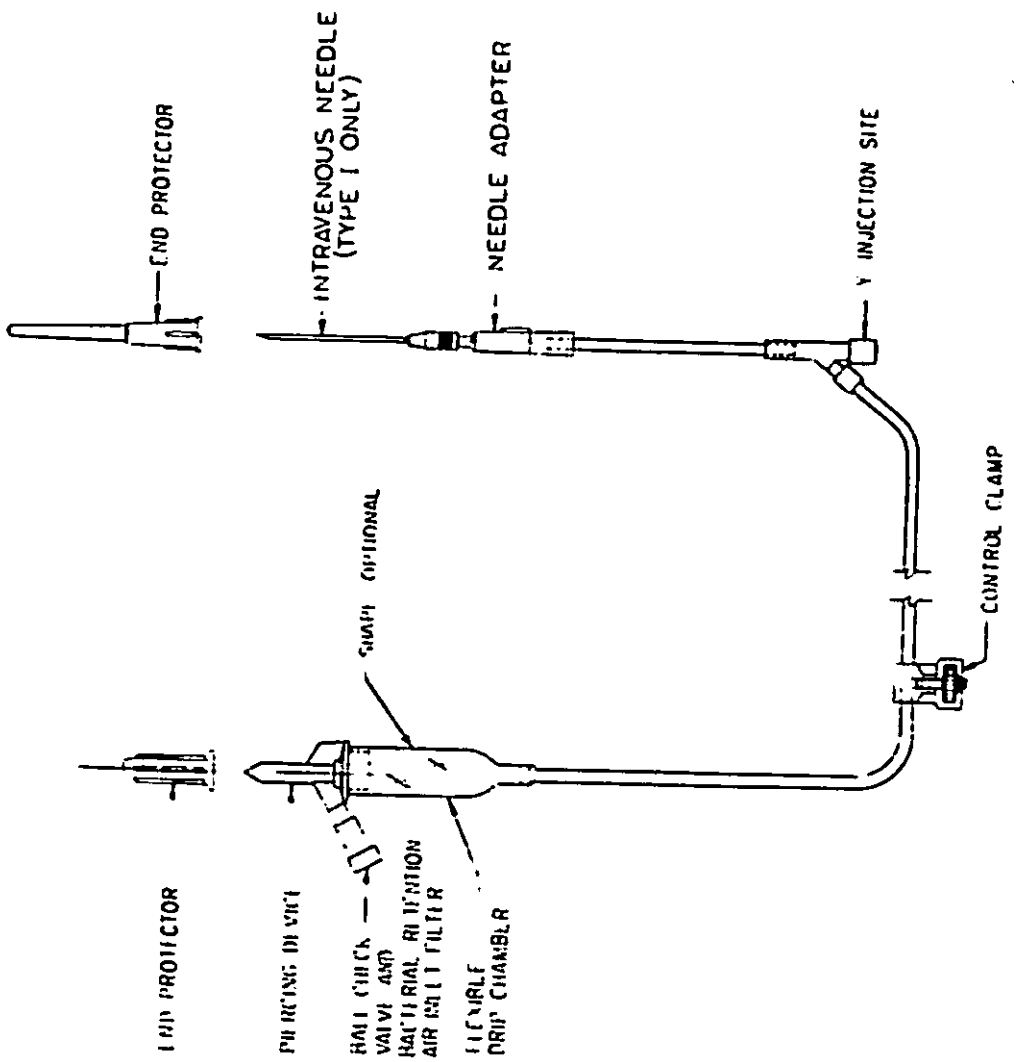
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency -DM

Project No. 6515-1756

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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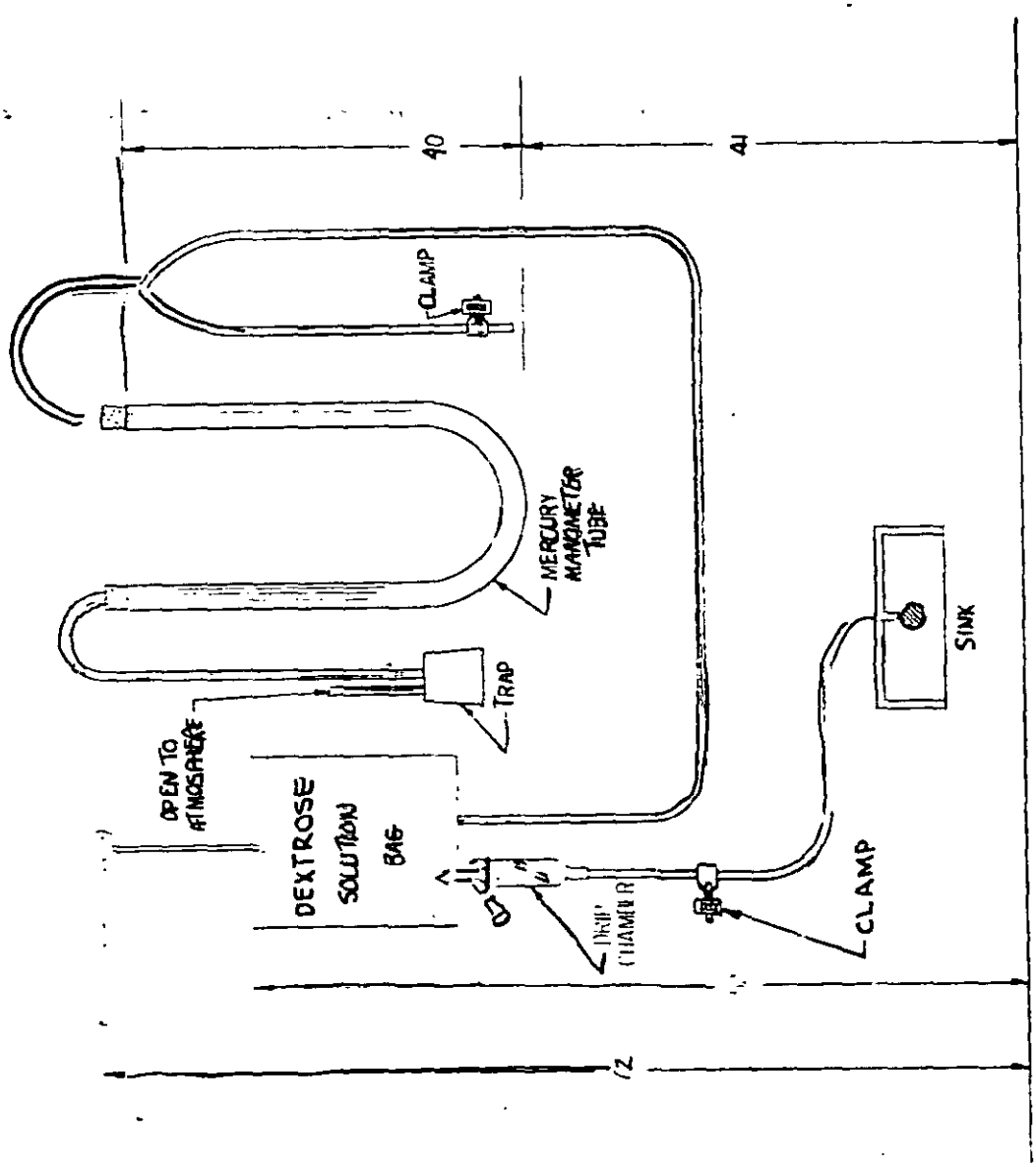


INTRAVENOUS INJECTION SET

FIGURE 1

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NOTE:
DIMENSIONS ARE IN INCHES AND ARE SUGGESTED ONLY.

FIGURE 2
LEAKAGE AND DRIP RATE SET-UP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-36996

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

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